

01)

```
/*
```

```
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change  
this license
```

```
 * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this  
template
```

```
*/
```

```
package com.mycompany.lab04;
```

```
/**
```

```
 *
```

```
 * @author LENOVO
```

```
 */
```

```
abstract class StaffMember {
```

```
    private String fullName;
```

```
    private final String staffId;
```

```
    protected String department;
```

```
    private static int staffCount = 0;
```

```
    public StaffMember(String fullName, String staffId, String department) {
```

```
        this.fullName = fullName;
```

```
        this.staffId = staffId;
```

```
        this.department = department;
```

```
        staffCount++; // count objects
```

```
    }
```

```
public String getFullName() {  
    return fullName;  
}
```

```
public String getStaffId() {  
    return staffId;  
}
```

```
public String getDepartment() {  
    return department;  
}
```

```
public final void displayBasicDetails() {  
    System.out.println("Name: " + fullName);  
    System.out.println("Staff ID: " + staffId);  
    System.out.println("Department: " + department);  
}
```

```
public abstract double calculateMonthlyPayment();
```

```
public static void showSystemName() {  
    System.out.println("Campus Staff Payment System");  
}
```

```
public static int getStaffCount() {
```

```

        return staffCount;
    }

    public void changeDepartment(String newDepartment) {
        if (newDepartment != null && !newDepartment.isEmpty()) {
            department = newDepartment;
        }
    }

    public final void showCommonNotice() {
        System.out.println("Notice: Salary is processed at end of each month.");
    }
}

/*
StaffMember is abstract because it provides a general design and should not be
instantiated directly.
*/

02)

/*
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*/

package com.mycompany.lab04;

```

```
/**
 *
 * @author LENOVO
 */
class Lecturer extends StaffMember {

    private int courseCount;
    private double paymentPerCourse;

    public Lecturer(String fullName, String staffId, String department,
        int courseCount, double paymentPerCourse) {

        super(fullName, staffId, department);
        this.courseCount = courseCount;
        this.paymentPerCourse = paymentPerCourse;
    }

    @Override
    public double calculateMonthlyPayment() {
        return courseCount * paymentPerCourse;
    }

    public void displayLecturerDetails() {
        displayBasicDetails();
        System.out.println("Courses: " + courseCount);
    }
}
```

```
        System.out.println("Payment per Course: " + paymentPerCourse);
    }
}
```

03)

```
/*
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 this license
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 template
 */
package com.mycompany.lab04;
```

```
/**
 *
 * @author LENOVO
 */
class LabAssistant extends StaffMember {

    private int hoursWorked;
    private double hourlyRate;

    public LabAssistant(String fullName, String staffId, String department,
        int hoursWorked, double hourlyRate) {

        super(fullName, staffId, department);
        this.hoursWorked = hoursWorked;
    }
}
```

```
    this.hourlyRate = hourlyRate;
}
```

```
@Override
```

```
public double calculateMonthlyPayment() {
    return hoursWorked * hourlyRate;
}
```

```
public void displayLabAssistantDetails() {
    displayBasicDetails();
    System.out.println("Hours Worked: " + hoursWorked);
    System.out.println("Hourly Rate: " + hourlyRate);
}
}
```

```
/*
department can be used because it is protected in the parent class.
*/
```

04)

```
/*
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* Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this
template
*/
```

```
package com.mycompany.lab04;
```

```

/**
 *
 * @author LENOVO
 */
final class UniversityPolicy {

    public static final String UNIVERSITY_NAME = "University of Kelaniya";
    public static final double BONUS_RATE = 0.10;

    public static void showPolicyHeader() {
        System.out.println("University Payment Policy");
    }

    public static double calculateBonus(double monthlyPayment) {
        return monthlyPayment * BONUS_RATE;
    }
}

/*
A final class cannot be inherited to prevent modification of important rules.
*/

```

05,06)

```

/*

```

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*/

package com.mycompany.lab04;

/**

*

* @author LENOVO

*/

public class Main {

public static void main(String[] args) {

StaffMember.showSystemName();

UniversityPolicy.showPolicyHeader();

System.out.println("University: " + UniversityPolicy.UNIVERSITY_NAME);

Lecturer lec1 = new Lecturer("Nimal Perera", "L001", "IT", 2, 50000);

Lecturer lec2 = new Lecturer("Kasuni Silva", "L002", "CS", 3, 45000);

LabAssistant lab1 = new LabAssistant("Saman Kumara", "A001", "IT", 120, 500);

// Change department

lec1.changeDepartment("Software Engineering");


```
System.out.println("\n--- Lecturer 1 ---");  
lec1.displayLecturerDetails();  
double pay1 = lec1.calculateMonthlyPayment();  
System.out.println("Monthly Payment: " + pay1);
```

```
System.out.println("\nLecturer 2");  
lec2.displayLecturerDetails();  
double pay2 = lec2.calculateMonthlyPayment();  
System.out.println("Monthly Payment: " + pay2);
```

```
System.out.println("\nLab Assistant");  
lab1.displayLabAssistantDetails();  
double pay3 = lab1.calculateMonthlyPayment();  
System.out.println("Monthly Payment: " + pay3);
```

```
double total = pay1 + pay2 + pay3;  
System.out.println("\nTotal Payment: " + total);
```

```
System.out.println("\nTotal Staff Count: " + StaffMember.getStaffCount());
```

```
System.out.println("\nNotices");  
lec1.showCommonNotice();  
lec2.showCommonNotice();  
lab1.showCommonNotice();
```

```
// Bonus
```

```
System.out.println("\nBonus");

System.out.println("Lecturer1 Bonus: " + UniversityPolicy.calculateBonus(pay1));

System.out.println("Lecturer2 Bonus: " + UniversityPolicy.calculateBonus(pay2));

System.out.println("Lab Assistant Bonus: " + UniversityPolicy.calculateBonus(pay3));

}

}

/*

changeDepartment() is useful because it controls how updates happen instead of allowing
direct access.

*/
```

```
--- exec:3.5.1:exec (default-cli) @ Lab04 ---
Campus Staff Payment System
University Payment Policy
University: University of Kelaniya

--- Lecturer 1 ---
Name: Nimal Perera
Staff ID: L001
Department: Software Engineering
Courses: 2
Payment per Course: 50000.0
Monthly Payment: 100000.0

Lecturer 2
Name: Kasuni Silva
Staff ID: L002
Department: CS
Courses: 3
Payment per Course: 45000.0
Monthly Payment: 135000.0

Lab Assistant
Name: Saman Kumara
Staff ID: A001
Department: IT
Hours Worked: 120
Hourly Rate: 500.0
Monthly Payment: 60000.0

Total Payment: 295000.0

Total Staff Count: 3

Notices
Notice: Salary is processed at end of each month.
Notice: Salary is processed at end of each month.
Notice: Salary is processed at end of each month.

Bonus
Lecturer1 Bonus: 10000.0
Lecturer2 Bonus: 13500.0
Lab Assistant Bonus: 6000.0

-----
BUILD SUCCESS
-----

Total time: 1.030 s
Finished at: 2026-04-17T22:59:45+05:30
-----
```